

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
ACADEMIC YEAR 2026 – 27

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COURSE: BFSI (Online)

SEMESTER : 3rd Class : Sy.Btech



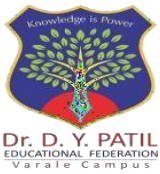
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CREDIT ASSISTANT: AI-Powered Credit Health Analysis & Financial Guidance Platform

Project Description :-

Credit Assistant is an AI-driven financial platform designed to help users understand their credit health, analyze their financial situation and receive personalized guidance for improving their credit profile. The platform focuses on helping Indian users make informed financial decisions by analyzing important financial factors such as credit score, income, expenses, outstanding debt, credit utilization and missed payments.

The system combines financial calculations with Generative AI to provide personalized financial insights and recommendations. It helps users identify factors that may affect their creditworthiness, understand their financial position and receive actionable suggestions for improving their credit profile.



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The application provides an interactive dashboard that presents financial information, credit-related indicators, progress visualizations and improvement recommendations. Its AI-powered financial assistant uses the Google Gemini API to generate explanations and guidance based on the user's financial information, making complex financial concepts easier to understand.

Credit Assistant is developed using a modern web application architecture with a React-based frontend, backend services, database integration and AI capabilities. The platform aims to make credit awareness and financial guidance more accessible through an interactive and user-friendly interface.

Technology & AI :-

Frontend:	React	/	Web-Based	User	Interface
Backend:	Server-Side	Application	/	Financial	Calculation
Database:	Database	Integration	for	Application	Data
AI:	Google	Gemini	API	/	Generative
Financial	Analytics:	Credit	Utilization	/	Debt-to-Income
Data	Visualization:	Interactive	Financial	/	Charts
Development	Tools:		Git	/	GitHub
Deployment:	Manus-hosted Web Application				

Key Features :-

- AI-powered financial guidance and credit health analysis
- Credit profile assessment using user-provided financial information
- Analysis of credit score and factors affecting credit health
- Income and expense analysis for understanding financial stability
- Credit utilization analysis to identify potential improvement areas
- Debt assessment and debt-to-income ratio calculations
- Identification of missed payments and other credit-related concerns
- Personalized financial recommendations generated using Google Gemini
- Interactive dashboard displaying important financial indicators
- Visual representation of financial progress and improvement areas
- Personalized roadmap to help users improve their credit profile
- Conversational AI assistant for credit-related and financial questions
- User-friendly interface for accessing financial insights and recommendations

Scenarios :-

Scenario 1: Credit Health Assessment

A user wants to understand their current credit health. The user enters relevant financial information, including their credit score, income, expenses and outstanding debts. Credit Assistant analyzes the available information, identifies important financial indicators and presents insights that help the user understand their current financial position.

Scenario 2: Credit Utilization and Debt Analysis

A user has outstanding debt and wants to understand its impact on their financial profile. The system analyzes the user's debt and credit utilization information, calculates relevant financial ratios where applicable and identifies areas that may require attention. The platform provides guidance to help the user understand possible steps for improving their financial situation.

Scenario 3: AI-Powered Financial Guidance

A user wants personalized suggestions for improving their credit profile. The user interacts with the AI financial assistant and provides relevant financial details or asks a credit-related question. The Google Gemini API generates contextual explanations and recommendations, helping the user understand financial concepts and possible improvement strategies.

Scenario 4: Financial Dashboard and Improvement Roadmap

A user opens the Credit Assistant dashboard to review their financial position. The dashboard presents available credit and financial indicators, visual summaries and identified improvement areas. The user can use these insights and recommendations to understand their progress and plan future financial improvements.

Milestones :-

Milestone 1 – Project Planning & Requirement Analysis

Identify the problem of limited credit awareness and access to personalized financial guidance. Define the project's objectives, target users, key financial indicators, application requirements and overall workflow.

Milestone 2 – Frontend Development & User Interface Design

Develop the React-based user interface and organize the application into relevant sections. Design an interactive dashboard that presents financial information, credit indicators and recommendations in an understandable format.

Milestone 3 – Backend Development & Financial Analysis

Develop backend functionality to process user-provided financial information. Implement relevant financial calculations, including debt-to-income ratio and credit utilization analysis, to identify important financial indicators and potential improvement areas.

Milestone 4 – AI Integration & Personalized Recommendations

Integrate the Google Gemini API to provide AI-generated financial explanations and recommendations. Connect the AI functionality with the application workflow so users can receive relevant guidance about credit health and financial management.

Milestone 5 – Dashboard, Data Visualization & Application Integration

Integrate the frontend, backend, financial analysis and AI assistant. Present financial indicators and improvement insights through the dashboard, charts and progress visualizations to make the results easier to understand.

Milestone 6 – Testing, Optimization & Deployment

Test the application's main workflows, financial calculations, dashboard components and AI integration. Identify and resolve implementation issues, improve usability and deploy the web application for online access. Maintain the source code using Git and GitHub.

Conclusion :-

Credit Assistant provides an AI-powered approach to understanding credit health and improving financial awareness. By combining financial analysis, relevant calculations, interactive visualizations and Generative AI, the platform helps users understand their financial situation and identify areas where improvement may be possible.

The project demonstrates the practical application of web development, backend programming, database integration, financial analytics and AI-based assistance in the Banking, Financial Services and Insurance (BFSI) domain. It illustrates how AI can help make financial information more understandable and support informed financial planning.



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Expected Outcome :-

Credit Assistant aims to help users develop a better understanding of their credit profile, financial obligations and credit-related improvement opportunities. The platform provides financial indicators, personalized AI-generated guidance and an interactive dashboard to support informed financial decision-making.

The project demonstrates practical knowledge of React-based frontend development, backend application architecture, database integration, financial calculations, Generative AI, Google Gemini API integration, data visualization, Git, GitHub and web application deployment. It also provides a foundation for future enhancements, including more comprehensive financial analytics, additional financial planning tools and integrations with authorized financial data sources.